Energy-Adaptive Hybrid Event-Driven Spoofing Detection (IAH-EDSD) in Industrial Wireless Sensor Networks

Master PhD Viva Preparation & Comprehensive Defense Package

Complete Technical Documentation, Derivations, System Architecture, Results Analysis, and Viva Q&A Defense Strategy

1st Prashanth K
Department of Master of Computer Applications
RV College of Engineering (Affiliated to VTU, Belagavi), Bengaluru, India
prashanthk@rvce.edu.in

2nd Usha J
Department of Master of Computer Applications
RV College of Engineering (Affiliated to VTU, Belagavi), Bengaluru, India
ushaj@rvce.edu.in

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Contents

I Master Thesis Chapter **2**

1 Energy-Adaptive Hybrid Event-Driven Spoofing Detection (IAH-EDSD) in Industrial WSNs **3**
1.1 Introduction . . . 3
1.1.1 Problem Statement . . . 3
1.1.2 Motivation . . . 3
1.1.3 Research Gap & Key Contributions . . . 3
1.2 Related Work & Taxonomic Positioning . . . 4
1.3 System Model & Operational Hypotheses . . . 4
1.3.1 Network Topology Model . . . 4
1.3.2 Radio Channel & Physical-Layer Model . . . 5

II Mathematical Formulation & Step-by-Step Derivation **6**

2 Mathematical Derivation Guide **7**
2.1 Spatial RSSI Anomaly Derivation . . . 7
2.2 Dynamic Multi-Factor Trust Accumulation Engine . . . 7

III Formal Algorithm Design & Computational Complexity **8**

3 Formal Algorithm & Complexity Analysis **9**
3.1 Algorithm Specification . . . 9
3.2 Computational Complexity Analysis . . . 9

IV Multi-Layer System Architecture & Defense **10**

4 System Architecture & Defense Strategy **11**
4.1 Multi-Layer Architecture . . . 11

V Analytical Results & Reviewer Discussion **12**

5 Analytical Results & Benchmark Discussion **13**
5.1 Master Results Summary . . . 13
5.2 Why Results Occur . . . 13

VI PhD Viva Defense Questions & Answers Guide 14

6 PhD Viva Defense Questions & Answers 15
6.1 Conceptual & Architectural Defense 15
6.2 Mathematical & Theoretical Defense 15

VII Viva Quick-Reference Exam Cheat Sheet 16

7 Viva Quick-Reference Exam Cheat Sheet 17
7.1 The 30-Second Elevator Pitch 17
7.2 Key Metric Reference Table 17

Part I

Master Thesis Chapter

Chapter 1

Energy-Adaptive Hybrid Event-Driven Spoofing Detection (IAH-EDSD) in Industrial WSNs

1.1 Introduction

1.1.1 Problem Statement

Industrial Wireless Sensor Networks (IWSNs) form the operational backbone of modern industrial automation, smart manufacturing, oil and gas pipelines, power grid monitoring, and Industry 4.0 applications. Operating under severe energy, bandwidth, and computational constraints, sensor nodes routinely transmit critical telemetry data back to centralized control systems or Base Stations (BS). However, due to the open, broadcast nature of wireless radio channels (e.g., IEEE 802.15.4 / ZigBee) and unencrypted header fields, IWSNs are fundamentally vulnerable to **active identity spoofing** and **Sybil attacks**. In these attacks, malicious adversaries forge MAC/IP source addresses or masquerade under multiple stolen node identities to inject false measurement data, trigger unnecessary industrial shutdowns, bypass access control protocols, or exhaust network routing resources.

1.1.2 Motivation

Traditional cryptographic mechanisms—such as public-key cryptography using Elliptic Curve Digital Signature Algorithm (ECDSA NIST P-256)—provide unforgeable per-packet origin authentication. However, executing an ECDSA-P256 signature verification on a resource-constrained 16-bit microcontroller (e.g., Texas Instruments MSP430 running at 8 MHz with < 10 KB RAM) incurs a heavy energy penalty of approximately 90.00 mJ per packet. In an industrial network carrying over 1.12 million telemetry frames, applying continuous cryptographic verification rapidly exhausts node initial energy reserves (e.g., 5.0 J for leaf nodes, 50.0 J for Cluster Heads), causing catastrophic network partition within hours.

Conversely, lightweight physical-layer authentication techniques (e.g., Received Signal Strength Indicator, RSSI) or purely behavioral anomaly detectors require minimal computation (5–100 μJ/packet), but suffer from high false positive rates (2.8–5.4%) under multipath fading, shadow fading, and dynamic obstacle movements typical of industrial factory floors.

1.1.3 Research Gap & Key Contributions

Existing research presents a rigid trade-off between **high security (heavy energy overhead)** and **low energy (weak security under dynamic threats)**. There is a distinct absence of

an event-driven, energy-aware security architecture that dynamically adapts its computational intensity to both instantaneous threat levels and residual battery reserves.

To bridge this research gap, this thesis introduces the **Industrial Adaptive Hybrid Event-Driven Spoofing Detection (IAH-EDSD)** framework. The principal contributions are:

1. **Event-Driven Hierarchical Triggers**: Replaces static continuous verification with an event-driven monitoring cascade based on physical-layer spatial RSSI deviation ($\Delta\text{RSSI}_i > 5.0$ dBm) and telemetry packet rate bursts ($\Phi(p_k)$), bypassing heavy audits during steady-state normal operations.

2. **Adaptive Multi-Factor Trust Engine**: Formulates a dynamic trust accumulation model ($T = \alpha T_{\text{auth}} + \beta T_{\text{behavior}} + \gamma T_{\text{energy}}$) featuring real-time context-aware weight updating and an Early Warning Score (EWS) temporal convergence filter.

3. **Three-Tier Energy-Adaptive Mode Switching**: Implements active energy-aware operational state selection (`FULL`, `LIGHT`, `EVENT_ONLY`) linked to residual battery capacity (E_{ratio}), guaranteeing microcontroller operational viability on 16-bit MSP430 MCUs.

4. **Empirical Industrial NS-3 Validation**: Validated via extensive discrete-event simulations across a 401-node industrial factory grid (1000 m $\times$ 1000 m) under $1,120,555$ telemetry packets, achieving 98.18% detection accuracy, 0.03% false positive rate, 99.40% per-packet energy reduction (0.540 mJ vs 90.00 mJ), and extending Cluster Head lifetime by over 130$\times$.

1.2 Related Work & Taxonomic Positioning

A critical analysis of state-of-the-art (SOTA) WSN intrusion detection and authentication mechanisms reveals four distinct paradigms:

Table 1.1: Taxonomic Comparison of WSN Spoofing Detection Paradigms.

Defense Paradigm	Key Mechanisms	Accuracy	Energy Overhead	Primary Limitatio
1. Continuous Cryptographic	ECDSA-P256, RSA	100.0%	90.00 mJ/pkt	Battery exhaustion i
2. Physical-Layer (PHY)	RSSI Fingerprinting	88.5%	0.005 mJ/pkt	High FPR under mul
3. Machine Learning (ML)	SVM, Random Forest	94.2%	0.100 mJ/pkt	Static non-adaptive I
4. Dynamic Trust Models	Beta / Fuzzy Trust	92.5%	0.055 mJ/pkt	Slow convergence on
5. Proposed IAH-EDSD	**Event + 3-Tier Hybrid**	**98.18%**	**0.540 mJ/pkt**	**Optimal Pareto Fr**

1.3 System Model & Operational Hypotheses

1.3.1 Network Topology Model

Consider an industrial WSN deployed over a square operational region $\mathcal{A} = [0, L] \times [0, L] \subset \mathbb{R}^2$ with $L = 1000$ m. The network consists of $N = 401$ physical nodes partitioned hierarchically:

- **Base Station (BS / Sink)**: Located at center coordinates $(500.0, 500.0)$ m with unlimited energy supply ($E_{\text{BS}} = 1000.0$ J).
- **Cluster Heads (CHs)**: $N_{\text{CH}} = 40$ nodes arranged in an 8×5 uniform grid. Initial energy $E_{\text{CH}}^{(0)} = 50.0$ J.
- **Leaf Sensor Nodes**: $N_{\text{leaf}} = 359$ nodes uniformly distributed within cluster boundaries. Initial energy $E_{\text{leaf}}^{(0)} = 5.0$ J.

1.3.2 Radio Channel & Physical-Layer Model

Transmissions adhere to IEEE 802.15.4 PHY/MAC standards on the 2.4 GHz ISM band operating via CC2420 RF transceivers ($P_{\text{tx}} = 0.0$ dBm $= 1.0$ mW, Data Rate $= 250$ kbps). Path loss follows the Log-Normal Shadowing Model:

$$P_{\text{rx}}(d) = P_{\text{tx}} - P_{\text{L}}(d_0) - 10n \log_{10}\left(\frac{d}{d_0}\right) + X_\sigma \tag{1.1}$$

where $P_{\text{L}}(d_0) = 40.0$ dB at reference $d_0 = 1.0$ m, $n = 2.7$, and $X_\sigma \sim \mathcal{N}(0, \sigma^2)$ with $\sigma = 2.0$ dB.

Part II

Mathematical Formulation & Step-by-Step Derivation

Chapter 2

Mathematical Derivation Guide

2.1 Spatial RSSI Anomaly Derivation

For a node i transmitting from coordinates $\boldsymbol{p}_i$, the expected Received Signal Strength at Cluster Head j at distance $d_{i,j} = \|\boldsymbol{p}_i - \boldsymbol{p}_j\|_2$ is:

$$\overline{\mathrm{RSSI}}_{\mathrm{exp}}^{(i,j)} = \mathbb{E}[P_{\mathrm{rx}}(d_{i,j})] = P_{\mathrm{tx}} - 40.0 - 27.0 \log_{10}\left(\|\boldsymbol{p}_i - \boldsymbol{p}_j\|_2\right) \tag{2.1}$$

When an adversary located at physical coordinates $\boldsymbol{p}_{\mathrm{att}} \neq \boldsymbol{p}_i$ transmits a spoofed packet claiming identity i, the actual physical transmission distance to CH j is $d_{\mathrm{att},j} = \|\boldsymbol{p}_{\mathrm{att}} - \boldsymbol{p}_j\|_2$.

The observed RSSI at CH j is:

$$\mathrm{RSSI}_{\mathrm{obs}}^{(i,j)}(t) = P_{\mathrm{tx}} - 40.0 - 27.0 \log_{10}\left(\|\boldsymbol{p}_{\mathrm{att}} - \boldsymbol{p}_j\|_2\right) + X_\sigma(t) \tag{2.2}$$

The spatial RSSI deviation metric $\Delta\mathrm{RSSI}_i(t)$ is:

$$\Delta\mathrm{RSSI}_i(t) = \left|\mathrm{RSSI}_{\mathrm{obs}}^{(i,j)}(t) - \overline{\mathrm{RSSI}}_{\mathrm{exp}}^{(i,j)}\right| = \left|27.0 \log_{10}\left(\frac{\|\boldsymbol{p}_i - \boldsymbol{p}_j\|_2}{\|\boldsymbol{p}_{\mathrm{att}} - \boldsymbol{p}_j\|_2}\right) + X_\sigma(t)\right| \tag{2.3}$$

2.2 Dynamic Multi-Factor Trust Accumulation Engine

Total node trust $T_i(t) \in [0, 1]$ at time t is computed via a weighted linear combination of three normalized trust metrics:

$$T_i(t) = \alpha(t) T_{\mathrm{auth}}^{(i)}(t) + \beta(t) T_{\mathrm{behavior}}^{(i)}(t) + \gamma(t) T_{\mathrm{energy}}^{(i)}(t) \tag{2.4}$$

subject to normalization constraint $\alpha(t) + \beta(t) + \gamma(t) = 1.0$.

Dynamic weights update adaptively based on temporal threat history (EWS_i):

$$\alpha(t) = \alpha_0 + \kappa \cdot \mathrm{EWS}_i(t), \quad \beta(t) = \beta_0 \cdot \left(1.0 - \frac{\sigma_{\mathrm{RSSI}}^2}{\sigma_{\max}^2}\right), \quad \gamma(t) = 1.0 - (\alpha(t) + \beta(t)) \tag{2.5}$$

where base parameters are $\alpha_0 = 0.40, \beta_0 = 0.40, \gamma_0 = 0.20$, and amplification factor $\kappa = 0.30$.

Part III

Formal Algorithm Design & Computational Complexity

Chapter 3

Formal Algorithm & Complexity Analysis

3.1 Algorithm Specification

Algorithm 1: IAH-EDSD Framework Execution Protocol

Input: Packet p_k, claimed sender ID i, RSSI RSSI_{obs}, residual energy $E_{\text{rem}}^{(j)}$
Output: Isolation verdict $\text{Isolate}_i \in \{0, 1\}$, trust $T_i(t)$, EWS score $\text{EWS}_i(t)$

1. Compute spatial RSSI deviation: $\Delta\text{RSSI}_i \leftarrow |\text{RSSI}_{\text{obs}} - \text{RSSI}_{\text{exp}}^{(i,j)}|$
2. Extract sequence gap ΔS_i and packet rate $R_i(t)$
3. Determine mode $\mathcal{M}$: FULL if $E_{\text{ratio}} > 0.70$, LIGHT if $0.30 < E_{\text{ratio}} \leq 0.70$, else EVENT_ONLY
4. **if** $\mathcal{M} ==$ EVENT_ONLY **then** process 1 in 3 packets; update trust; **return** verdict
5. Evaluate Event Trigger $\Phi(p_k) = \mathbb{I}(\Delta\text{RSSI}_i > 5.0 \vee \Delta S_i > 1 \vee R_i > 10)$
6. **if** $\Phi(p_k) == 0$ **then return** $\text{Isolate}_i \leftarrow 0$ (Steady-state bypass at $5\,\mu$J)
7. Update dynamic weights $\alpha(t), \beta(t), \gamma(t)$
8. Calculate $T_{\text{behavior}}^{(i)}(t)$ and $T_{\text{energy}}^{(i)}(t)$
9. **if** $\mathcal{M} ==$ FULL **and** $T_i(t-1) < 0.60$ **then** execute ECDSA-P256 audit (90 mJ)
10. Compute composite trust $T_i(t) = \alpha T_{\text{auth}} + \beta T_{\text{behavior}} + \gamma T_{\text{energy}}$
11. Update sliding window $\text{EWS}_i(t)$ over past $W = 50$ frames
12. **if** $T_i(t) < 0.60 \wedge \text{EWS}_i(t) > 0.40 \wedge \Delta\text{RSSI}_i > 5.0$ dBm **then** $\text{Isolate}_i \leftarrow 1$ **else** $\text{Isolate}_i \leftarrow 0$
13. **return** $\text{Isolate}_i, T_i(t), \text{EWS}_i(t)$

Figure 3.1: Algorithmic execution procedure of IAH-EDSD.

3.2 Computational Complexity Analysis

- **Time Complexity**: Steady-state screening takes $\mathcal{O}(1)$ scalar operations (1.2 μs). Sliding window update takes $\mathcal{O}(W) = \mathcal{O}(50) = \mathcal{O}(1)$. Per-packet time complexity is bounded constant time $\mathcal{O}(1)$.
- **RAM Footprint**: State vector per neighbor requires 43 bytes. For 15 neighbors, RAM footprint is 645 bytes $<$ 1.0 KB, fitting within 16-bit TI MSP430 RAM (10 KB).

Part IV

Multi-Layer System Architecture & Defense

Chapter 4

System Architecture & Defense Strategy

4.1 Multi-Layer Architecture

- **Layer 1: Leaf Sensor Nodes ($N_{\mathbf{leaf}} = 359$)**: TI MSP430 MCU, CC2420 Transceiver, 5.0 J battery. Transmits telemetry frames without running local detection to save power.
- **Layer 2: Cluster Heads ($N_{\mathbf{CH}} = 40$)**: 8×5 spatial grid, 50.0 J initial energy. Runs event trigger, dynamic trust engine, and targeted ECDSA audits.
- **Layer 3: Base Station Sink ($N_{\mathbf{BS}} = 1$)**: Center coordinates $(500, 500)$, 1000.0 J energy. Maintains global topology and blacklists.

Part V

Analytical Results & Reviewer Discussion

Chapter 5

Analytical Results & Benchmark Discussion

5.1 Master Results Summary

Table 5.1: Verified Experimental Performance Summary ($1,120,555$ Packets).

Performance Metric	**Measured Empirical Value**
Total Evaluated Telemetry Workload	1,120,555 Packets
Spoofing Detection Rate (TPR)	98.18%
False Positive Rate (FPR)	0.03%
Cryptographic Audit Activation Rate	0.53% (99.47% Bypass)
Average Evaluation Energy per Packet	0.540 mJ / packet
Continuous ECDSA Baseline Energy	90.00 mJ / packet
Per-Packet Energy Reduction	99.40% Reduction
Cluster Head Network Operational Lifetime	151.3 Hours (vs 1.16 Hours)
Network Lifetime Extension Factor	¿ 130x Extension

5.2 Why Results Occur

The 99.40% energy reduction occurs because 99.47% of benign steady-state packets trigger the fast event bypass ($\Phi = 0$), costing only $5\,\mu$J (0.005 mJ). Heavy ECDSA verification (90 mJ) is invoked on fewer than 0.53% of frames, yielding an average cost of 0.540 mJ/packet.

Part VI

PhD Viva Defense Questions & Answers Guide

Chapter 6

PhD Viva Defense Questions & Answers

6.1 Conceptual & Architectural Defense

Q: Why use a 3-tier hybrid framework instead of a single deep machine learning model?
Answer: Deep neural networks require > 100 KB RAM and continuous floating-point matrix multiplication, which is impossible on a 16-bit MSP430 node with < 10 KB RAM. Furthermore, deep ML models consume constant high energy regardless of threat levels. IAH-EDSD takes only 645 bytes RAM and cuts average energy by 99.40%, making it hardware-feasible.

6.2 Mathematical & Theoretical Defense

Q: Explain how trust dynamic weights (α, β, γ) maintain normalization.
Answer: Dynamic weights are defined as $\alpha(t) = \alpha_0 + \kappa \cdot \mathrm{EWS}_i(t)$, $\beta(t) = \beta_0(1 - \sigma^2_{\mathrm{RSSI}}/\sigma^2_{\max})$, and $\gamma(t) = 1-(\alpha(t)+\beta(t))$. Summing them yields $\alpha+\beta+[1-(\alpha+\beta)] = 1.0 \quad \forall t$, mathematically guaranteeing normalization without iterative projection costs.

Part VII

Viva Quick-Reference Exam Cheat Sheet

Chapter 7

Viva Quick-Reference Exam Cheat Sheet

7.1 The 30-Second Elevator Pitch

"Industrial WSNs are vulnerable to identity spoofing. Continuous ECDSA signatures cost 90 *mJ per packet, killing Cluster Head batteries within 1.16 hours. We created IAH-EDSD—an Energy-Adaptive Event-Driven framework. IAH-EDSD replaces continuous crypto with physical-layer RSSI screening (*$5\,\mu J$*) and dynamic multi-factor trust (*$T = \alpha T_{auth} + \beta T_{behavior} + \gamma T_{energy}$*). Heavy ECDSA audits are triggered on only 0.53% of packets, cutting energy consumption by 99.40% (*0.540 *mJ vs* 90.00 *mJ), maintaining 98.18% detection accuracy with a 0.03% false positive rate, and extending network lifetime by over 130×."*

7.2 Key Metric Reference Table

Table 7.1: Viva Exam Metric Quick Reference.

Metric	**Value**	**Key Context**
Network Topology	401 Nodes	1 Sink + 40 CHs + 359 Leaf (1000 m × 1000 m)
Detection Rate (TPR)	98.18%	Vs 88.5% (Pure PHY), 92.5% (DynTrust)
False Positive Rate	0.03%	Vs 5.40% (Pure PHY), 1.85% (DynTrust)
Crypto Audit Rate	0.53%	99.47% Steady-State Bypass Rate
Per-Packet Energy	0.540 mJ	99.40% Reduction vs 90.00 mJ Baseline
CH Network Lifetime	151.3 Hours	¿130x Extension vs 1.16 Hours Baseline
MCU Target	TI MSP430	16-bit @ 8 MHz, ¡10 KB RAM (645 B footprint)